

Crisis Regime Analysis

Onset / Chronic / Recovery / Stable — Fact-Based Results

1. Regime Definitions

The regime of each test observation is determined by combining the current outcome (crisis or not) with whether the same district was already in crisis the prior period (`ipc_lag_1`). This is a post-hoc diagnostic label — it is not used during training, only for interpreting results.

Onset — crisis now, no crisis prior: a district tipping into crisis for the first time

Chronic — crisis now, crisis also prior: a district locked in persistent crisis

Recovery — no crisis now, crisis prior: a district that just exited crisis

Stable — no crisis now, no crisis prior: a district with no recent crisis history

2. Test-Set Composition (2-year window, 7 folds, n=2,148)

Regime	Obs	Crisis obs	% of crises	Prevalence
Chronic	460	460	72.3%	100%
Onset	176	176	27.7%	100%
Recovery	98	0	—	0%
Stable	1,414	0	—	0%
Total	2,148	636	100%	29.6%

3. Chronic — History Is Enough

n = 460 crisis observations (72.3% of all crises). AR-Only recall: 100% AR+News recall: 100%. AR-Only median P(crisis): 0.795 AR+News median: 0.713

These are districts already in crisis last period. The model knows this through `ipc_lag_1`, which alone pushes the predicted probability well above 0.5 — no news is needed. Both models detect every chronic crisis in every fold. Adding news information actually lowers the predicted probability for 68.7% of chronic cases (median shift -0.082), because news coverage of a chronic crisis looks different from news coverage of an emerging one. The AR margin is wide enough that this makes no difference to detection — but it illustrates the models are using different signals.

4. Onset — Where News Earns Its Keep

n = 176 crisis observations (27.7% of all crises). AR-Only recall: 7.4% (13 of 176) AR+News recall: 25.6% (45 of 176) AR-Only median P(crisis): 0.165 AR+News median: 0.450

These are the hardest cases — a district falling into crisis for the first time, with no prior signal in the data. The AR model, having nothing in the history to flag, assigns a very low probability (90th percentile only 0.396). News fills the gap: reports of conflict, food insecurity, or displacement arrive before the IPC assessment does, lifting the median predicted probability from 0.165 to 0.450.

This is enough to push 45 of 176 onset crises over the detection threshold — though 74% remain undetected even with news.

Fold-by-fold onset breakdown:

Fold	n onset	AR recall	AR+News recall	Net saves
1	11	9%	55%	+5
2	2	0%	50%	+1
3	5	0%	0%	+0
4	26	0%	0%	+0
5	43	0%	23%	+10
6	53	23%	53%	+16
7	36	0%	0%	+0

All 32 net saves come from onset. News works in 4 of 7 folds; in the other 3, neither model detects a single onset crisis — the news signal was not strong enough.

5. Recovery — The AR Model Cannot Let Go

n = 98 observations, all non-crisis (target=0). AR-Only false alarm rate: 100% AR+News false alarm rate: 100%.
AR-Only median P(crisis): 0.807 AR+News median: 0.717

Recovery districts have just exited crisis — they are not in crisis now, but they were last period. The AR model sees `ipc_lag_1=1` and treats them identically to chronic crisis districts, assigning crisis-level confidence to all 98 of them (median 0.807). It has no way to distinguish 'was in crisis and still is' from 'was in crisis and recovered'. Adding news pulls the predicted probability down (median 0.807 → 0.717), suggesting the news landscape of a recovering district does look different — but not different enough to cross back below 0.5. Both models flag every recovery district as a false alarm.

6. Stable — News Is Sensitive to Quiet Districts

n = 1,414 observations, all non-crisis (target=0). AR-Only false alarm rate: 0% AR+News false alarm rate: 3.0% (43 cases).

With no prior crisis signal, the AR model is very conservative — it assigns low probabilities to all stable districts and produces zero false alarms. News changes this. The combined model raises the predicted probability for 86% of stable observations (median $\Delta = +0.200$), the largest and most consistent upward shift across all four regimes. This makes intuitive sense: news coverage of conflict, weather events, or food stress can be elevated in a district that has not yet reached IPC Phase 3. In 1,371 of 1,414 cases the model stays correctly below 0.5, but in 43 it crosses the threshold — generating false alarms in places where the IPC record shows no crisis, current or prior.

7. Overall False Alarm Rates (all non-crisis observations)

Total non-crisis observations: 1,512 (stable 1,414 + recovery 98)

AR-Only: 98 false alarms (6.5%) — all from recovery districts

AR+News: 141 false alarms (9.3%) — recovery (98) + stable (43)

AR-Only false alarms come entirely from its inability to recognise recovery — a structural limitation of the lag feature. AR+News inherits that problem and adds a new one: it raises probabilities in stable districts enough to trigger 43 additional false alarms.

The net result is 43 more false alarms than AR-Only, a trade-off against the 32 additional onset crises it correctly detects.

8. Net Balance

AR+News uniquely detects 32 crisis cases that AR-Only misses — all of them onset.

AR+News loses 0 detections that AR-Only makes — no regressions in chronic or any other regime.

Net saves: +32 (5.0% of 636 crisis observations). Both detect: 473. Neither detects: 131 (20.6%).

The intuitive story across regimes is consistent. For chronic crises, past history is the dominant signal and news is redundant. For onset crises, history is silent and news becomes the only available signal — imperfect, but real. For recovery districts, the lag feature misfires because it cannot distinguish exit from persistence; news nudges the model in the right direction but not far enough. For stable districts, news is sensitive to early-warning signals that may not yet be visible in IPC assessments — a potential strength, but one that also generates false alarms where no crisis materialises.